

AXIOM PROTOCOL

Axiom Technical Paper

Layer architecture, consensus flow, block structure, node roles, fees, AI accounts, and Pulse Layer 2.

Trust Mathematics, Not Power.

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Layered architecture

Axiom should be built as a layered system. Layer 1 is the conservative settlement and security layer. Pulse Layer 2 handles fast micropayments. Axiom Agent Layer provides permissioned AI accounts. Application Layer includes wallets, DeFi, ticketing, marketplaces, identity, and governance UI. Network & Storage handles peer-to-peer gossip, state sync, data availability, and light-client proofs.

Reference stack

Application Layer

- Wallets, DeFi, ticketing, marketplaces, AI agents, governance UI

Axiom Agent Layer

- Permissioned AI accounts, session keys, risk firewall, rate limits

Pulse Layer 2

- Micropayments, channels, rollups, AI-to-AI small payments

Axiom Core Layer 1

- Consensus, settlement, native AXM, smart contracts, finality

Network & Storage Layer

- P2P gossip, state sync, data availability, light-client proofs

Block structure

Block

- Header: chain_id, height, previous_block_hash, timestamp, epoch_id, proposer_id, randomness_seed, protocol_version
- Body: transaction_list, system_transactions, l2_settlement_batches
- Commitments: transaction_root, receipt_root, state_root, validator_set_root, data_availability_root
- Proofs: proposer_signature, aggregated_committee_signature, optional validity proofs, fraud proof references

The block format supports full nodes, light nodes, mobile proof verification, rollup settlement, and future validity proofs. State commitments allow devices to verify claims without downloading the entire chain.

Transaction lifecycle

- User, app, or AI agent creates a transaction from a smart account or standard account.
- Wallet simulates the transaction and displays risk warnings if needed.
- Transaction enters the mempool with adaptive base fee and optional priority fee.
- A proposer builds a block from valid transactions.
- Committee validators verify execution, state transition, signatures, fee rules, and policy constraints.

- If the required supermajority signs, the block receives soft finality.
- Checkpoint or BFT finality gives stronger finality after additional steps.
- Citizen Validators and Mobile Guardians continue to verify proofs and report anomalies.

Consensus: Proof of Integrity

Proof of Integrity is the proposed Axiom consensus family. Early testnets should implement it conservatively as proven PoS+BFT with anti-whale weighting and validator caps. Experimental reputation or mobile participation features should be added only after simulations and audits.

Validator selection score = $f(\text{sqrt}(\text{stake})) + \text{uptime_score} + \text{correctness_score} + \text{decentralization_bonus} + \text{long_term_reputation_penalty_score}$

Axiom should not use 1 coin = 1 unit of power. Square-root stake and caps reduce the marginal influence of large capital holders while still requiring economic commitment.

Node roles

- Guardian Core Node: produces blocks, executes smart contracts, signs consensus, and accepts slashing risk.
- Citizen Validator: verifies blocks and state transitions, monitors validators, and submits fraud proofs.
- Mobile Guardian: verifies headers, Merkle proofs, signatures, and fraud/validity proofs; it does not secure critical finality directly.
- Passive Witness: relays signals, helps report phishing/scams, and strengthens network availability with low-risk participation.

Why mobile is not a full validator

Mobile devices are powerful but not reliable enough to directly secure finality. They can lose power, network, storage, or be compromised by malware. Axiom uses mobile devices as light verifiers and fraud witnesses, not as the critical finality layer.

Axiom VM

Axiom VM should be EVM-compatible for adoption but stricter by default for safety-critical applications. It should support deterministic execution, gas metering, sandboxing, verified contract metadata, safe account standards, and optional formal verification for high-value contracts.

Adaptive market fee

Fees should not be fixed forever. They should respond to demand and supply of block space.

Total Fee = Base Fee + Priority Fee

Base Fee adjusts upward when block demand exceeds target utilization and downward when demand is below target utilization.

Priority Fee is an optional tip for faster inclusion. Layer 1 fees protect settlement from spam; Layer 2 fees support low-cost payments.

Axiom Agent Layer

AI agents should be first-class users, but they must not hold unlimited wallet authority. Agent accounts use session keys, spending limits, whitelists, risk policies, audit logs, rate limits, and emergency revocation.

Example policy: daily_limit 5 AXM; per_transaction_limit 0.05 AXM; allowed_network Pulse; allowed_contracts verified whitelist; expiration 24 hours; human approval above 0.10 AXM; owner can revoke immediately.

Pulse Layer 2

Pulse handles small payments for people, merchants, AI agents, games, IoT devices, APIs, content, and computation. It may start with payment channels and later support rollups.

Pulse payment flow:

1. User locks funds on Axiom L1.
2. User or AI agent performs many small Pulse payments.
3. Pulse periodically settles a compressed state or batch proof to L1.
4. L1 validators verify the settlement proof and collect settlement fees.

Disclaimer

This document is a Phase 0 research and design specification. It is not investment advice, legal advice, security assurance, or a promise that the system is feasible without further research, simulations, audits, and testnet validation.